

2022-Ph-H1-Q22

Section: Electricity

Topic: Current, PD, Power, Resistance

A 12 V battery is connected across a resistor and the current is 2 A. What is the resistance?

Answer: C

(6 Ω)

Guidance for Students:

Use Ohm's Law:

$$V = IR \Rightarrow R = \frac{V}{I} = \frac{12}{2} = 6 \Omega$$

Revision Tips:

- Rearranged Ohm's Law: $R = \frac{V}{I}$
- Always check units: volts, amps, ohms
- Watch for prefixes like k Ω or mA in other questions